

MANIITSOQ SUKKERTOPPEN, Greenland

WMO: 042410

Lat: 65.4110N

Lon: 52.9325W

Elev: 105

StdP: 14.64

Time Zone: -3.00 (GRL)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	-4.3	0.4	-14.6	2.5	-1.6	-10.7	3.1	4.2	44.0	27.1	36.2	18.1	11.0	80	0.906

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.4	59.5	48.7	56.5	47.7	54.0	46.8	50.6	56.2	49.2	54.3	48.0	52.4	9.8	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
48.0	49.8	51.1	46.5	46.9	49.6	45.3	44.8	48.7	20.7	56.2	19.9	54.6	19.2	52.6	57.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 34.8	29.2	25.1	-4.8	69.1	6.7	3.4	-9.6	71.5	-13.5	73.5	-17.2	75.4	-22.0	77.8		
(5)			-5.4	54.0	6.3	2.1	-10.0	55.5	-13.7	56.7	-17.3	57.9	-21.9	59.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	31.4	18.7	18.4	18.2	26.8	35.0	42.3	47.6	46.7	41.3	33.6	26.4	21.3		
	DBStd	12.85	8.60	10.44	10.55	7.47	5.80	4.80	4.93	3.86	3.98	5.88	6.64	7.94		
	HDD50	6831	970	885	985	695	465	234	107	118	264	511	707	890		
	HDD65	12250	1435	1305	1450	1145	929	680	540	567	712	975	1157	1355		
	CDD50	56	0	0	0	0	1	4	33	15	2	1	0	0		
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH74	1	0	0	0	0	0	0	1	0	0	0	0	0		
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	10.9	12.6	11.6	10.9	10.5	9.8	9.9	9.3	9.5	11.6	10.5	12.1	11.9		
	PrecAvg	23.2	2.2	1.4	1.4	1.4	1.3	1.4	1.9	2.6	2.6	2.2	2.5	2.4		
(15) Precipitation	PrecMax	32.4	6.2	3.4	3.6	3.2	2.3	3.0	3.6	4.3	5.6	4.6	5.3	5.0		
	PrecMin	15.4	0.2	0.6	0.1	0.5	0.3	0.5	0.7	0.9	0.7	0.6	0.9	1.0		
	PrecStd	4.2	1.4	0.7	0.8	0.7	0.6	0.6	0.8	0.9	1.0	0.9	1.1	1.0		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	40.7	45.0	45.7	49.2	54.9	60.1	64.5	63.4	56.4	52.6	46.5	41.9			
	0.4% MCWB	34.7	35.2	37.9	39.0	43.6	47.0	51.0	50.6	44.9	42.5	38.3	34.5			
	2% DB	37.2	40.7	38.9	42.9	49.5	55.4	60.4	57.7	51.9	46.5	41.3	38.5			
	2% MCWB	32.0	32.3	34.0	36.8	41.5	46.2	49.8	48.4	42.8	38.4	36.0	34.6			
	5% DB	34.4	36.6	35.7	39.3	45.8	52.3	57.5	54.7	48.9	42.9	37.7	35.5			
	5% MCWB	30.2	30.7	32.0	34.3	39.7	45.0	48.8	47.6	41.8	37.2	34.5	32.9			
	10% DB	31.5	32.1	32.7	36.3	42.8	49.4	55.0	52.4	46.6	40.4	35.3	32.1			
	10% MCWB	28.5	28.6	29.3	32.4	38.3	44.0	48.0	46.9	41.0	35.7	32.8	30.0			
	0.4% WB	37.3	36.6	39.1	41.1	44.8	50.3	53.7	52.4	48.5	44.0	41.5	37.2			
	0.4% MCDB	39.1	42.1	44.3	44.3	51.8	54.1	60.0	57.6	53.9	51.8	44.1	38.4			
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	33.2	33.7	34.9	37.6	42.5	47.1	51.2	50.2	45.0	40.2	37.7	35.4			
	2% MCDB	35.7	39.2	38.4	41.6	47.3	52.5	57.4	55.0	48.0	43.1	39.8	37.7			
	5% WB	30.5	31.0	32.3	34.9	40.5	45.6	49.8	48.9	43.5	38.2	34.8	32.9			
	5% MCDB	34.1	35.5	35.3	38.5	44.9	50.9	55.5	53.2	46.5	41.3	36.7	35.1			
	10% WB	28.1	28.4	29.5	32.9	38.4	44.3	48.5	47.8	42.1	36.6	32.7	30.2			
	10% MCDB	31.8	33.1	32.4	35.7	42.2	49.0	53.6	51.4	45.1	39.6	34.9	32.5			
(35) Mean Daily Temperature Range	MDBR	6.5	8.2	8.6	8.2	8.0	8.8	9.4	8.0	6.6	6.4	6.2	5.9			
	5% DB MCDBR	9.5	12.5	10.9	11.1	13.4	14.7	14.5	12.4	10.1	9.4	9.1	9.0			
	5% DB MCWBR	7.4	9.2	8.3	7.2	7.3	6.7	6.3	5.9	5.6	6.1	7.9	6.8			
	5% WB MCDBR	9.4	12.1	11.0	10.1	12.5	12.6	12.9	10.4	8.1	7.9	9.1	8.5			
	5% WB MCWBR	8.2	9.9	8.6	7.3	7.4	6.5	6.4	5.9	5.1	6.4	8.4	7.1			
(40) Clear-Sky Solar Irradiance	taub	N/A	0.238	0.267	0.320	0.331	0.343	0.341	0.340	0.303	0.259	0.356	0.350			
	taud	N/A	2.272	2.364	2.282	2.352	2.425	2.475	2.484	2.512	2.329	2.828	2.914			
	Ebn at Noon	N/A	217	263	269	277	276	273	262	249	201	102	56			
	Edh at Noon	N/A	15	23	32	33	31	29	26	20	14	6	4			
(44) All-Sky Solar Radiation	RadAvg	28	180	603	1184	1643	1797	1646	1199	712	303	57	7			
	RadStd	3	17	56	92	111	156	158	83	80	27	11	1			

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)		+1.31	N/A	N/A	+2.54	N/A	N/A	N/A	N/A	+20	N/A

Nomenclature: See separate page